Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 34)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

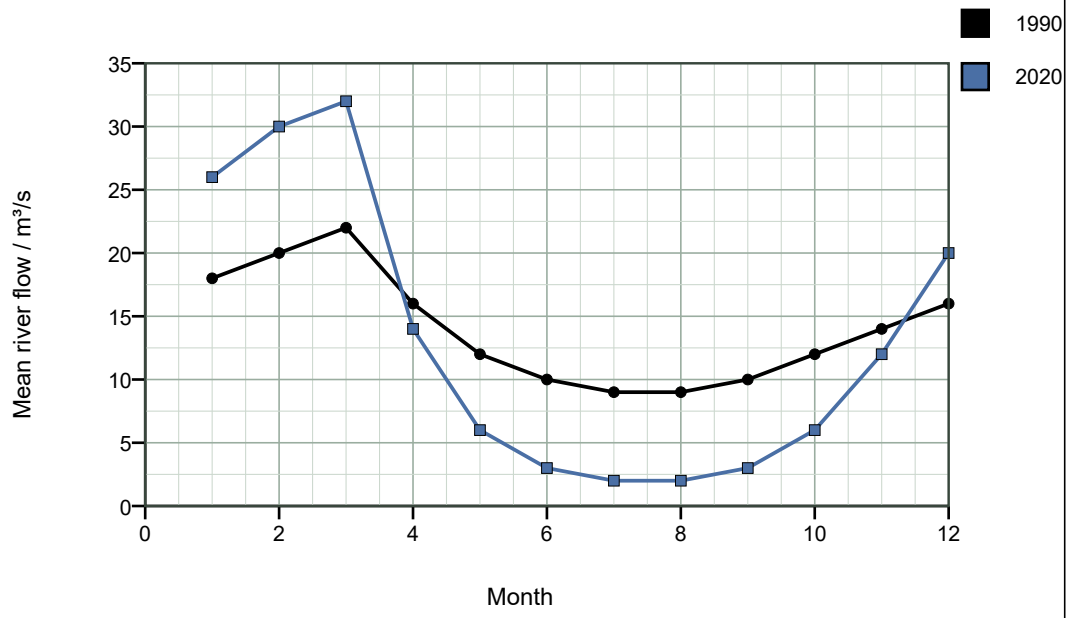
Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the monthly flow of a river in South America that is fed partly by glacier meltwater, in 1990 and 2020. The glacier feeding the river shrank considerably between the two dates.

Figure 1



0 1 . 1 Compare the river flow in 1990 and 2020.

[3 marks]

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0 1 . 2 Explain how the shrinking of the glacier explains the differences.

[3 marks]

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0 1 . 3 Suggest **two** problems the change in flow causes for people living downstream.

[2 marks]

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2 .....

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0 1 . 4 Explain why a glacier can be extending down its valley yet still be losing ice volume.

[2 marks]

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0 2 A mining company is assessing a copper deposit beneath farmland.

0 2 . 1 Describe what each of the following survey techniques detects.

[3 marks]

Gravimetry .....

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Magnetometry .....

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Seismic surveys .....

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0 2 . 2 Explain why trial drilling is normally carried out only after the remote sensing surveys.

[2 marks]

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0 2 . 3 Explain how a rise in the world price of copper can turn part of the **resource base** into a **reserve**.

[2 marks]

0 2 . 4 Explain **four** environmental impacts of extracting the copper by opencast mining.

[4 marks]

0 2 . 5 Describe how the site could be restored after closure, and explain why restoration is planned before mining begins.

[4 marks]

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15

0 3 Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

0 3 . 1 Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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0 3 . 2 Explain the purpose of aeration (stage 3).

[2 marks]

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0 3 . 3 Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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0 3 . 4 Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic) .....

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Salt .....

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Odours and pesticides .....

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0 4 Table 1 shows the concentration of cadmium in rice and in the kidneys of people living in two villages in Japan. Village A is downstream of a zinc mine.

Table 1

Village	Cadmium in rice / mg kg^{-1}	Cadmium in kidney of residents / mg kg^{-1}	Cases of bone disease (itai-itai) per 1000 people
A (downstream of mine)	1.2	180	34
B (control)	0.05	22	0

0 4 . 1 Calculate how many times more concentrated cadmium is in the kidneys of village A residents than in their rice.

[1 mark]

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Answer [ ] times

0 4 . 2 Explain why cadmium and zinc pollution often occur together.

[1 mark]

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0 4 . 3 Use Table 1 and your knowledge of the properties of cadmium to explain the high rate of bone disease in village A.

[4 marks]

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0 4 . 4 Suggest **two** ways cadmium pollution from the mine could be controlled.

[2 marks]

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2 .....

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0 4 . 5 Cadmium is used in cadmium-telluride photovoltaic panels. Suggest why this new use may cause pollution in the future.

[2 marks]

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0 5 Table 1 shows the mean surface temperature of the Earth with and without an atmosphere containing greenhouse gases.

Table 1

Condition	Mean surface temperature / °C
No greenhouse effect	-18
Natural greenhouse effect	15

0 5 . 1 Use Table 1 to calculate the warming produced by the natural greenhouse effect.

[1 mark]

Answer °C

0 5 . 2 Explain how the natural greenhouse effect warms the Earth's surface.

[4 marks]

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0 5 . 3 Explain what is meant by a **greenhouse gas**.

[1 mark]

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0 5 . 4 The wavelengths of radiation arriving at the Earth are different from those leaving. Describe this difference and explain why it matters to the energy budget.

[2 marks]

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0 5 . 5 Explain why a change in the movement of energy through the atmosphere could affect the distribution of species.

[2 marks]

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0 6 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 6 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 6 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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2 .....

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5

0 7 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 7 . 1 Calculate the estimated population size.

[2 marks]

Answer ..... beetles

0 7 . 2 State **four** assumptions of the Lincoln Index method.

[4 marks]

1 .....

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2 .....

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0 7 . 3 Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

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0 7 . 4 Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

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0 8 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 8 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 8 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

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0 9 . 1 Explain how affluence and the strategies of manufacturers increase the amount of solid waste produced.

[4 marks]

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0 9 . 2 Give **three** factors that affect the choice of disposal method for a solid waste.

[3 marks]

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2 .....

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3 .....

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0 9 . 3 Explain **one** environmental advantage and **one** disadvantage of composting garden and food waste rather than sending it to landfill.

[2 marks]

Advantage .....

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Disadvantage .....

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0 9 . 4 Explain why mixed municipal waste is difficult to recycle.

[1 mark]

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1 0 . 1 Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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Extra space

1 0 . 2 State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS